Xiaoxingkai Transfer Result Report

Run directory: S:\STU-Papers\My_Papers\EXP1-Pre-Training-Standalone\outputs\runs\xiaoxingkai_transfer_daily12_history_autoreg\xiaoxingkai_transfer_daily12_history_autoreg_20260415_133032
Experiment: xiaoxingkai_transfer_daily12_history_autoreg
Source file: S:\STU-Papers\My_Papers\EXP1-Pre-Training-Standalone\data\raw\LakeIce_ERA5_less60_XiaoxingkaiDaily12.xlsx
Target lake: [Li] Lake Xiaoxingkai

Front Summary

Split	Loss	RMSE	MAE	R2	Bias	Negative Preds
val	0.0115	0.1064	0.0961	0.8232	0.0634	0
test	0.0215	0.1466	0.1390	-11.6030	-0.1390	0

Per-Lake Numeric Evaluation Summary

Lake	Count	RMSE	MAE	R2	Bias
[Li] Lake Xiaoxingkai	36	0.1466	0.1390	-11.6030	-0.1390

Key Diagnostic Notes

Best epoch: 118

Observed evaluation mean/std: 0.7567 / 0.0413

Predicted evaluation mean/std: 0.6177 / 0.0640

Validation windows: 40 | Test windows: 36

Interpretation: if predicted std is much smaller than observed std, the model is behaving too conservatively and is under-expressing peak-to-trough variation.

0. Training Inputs And Hyperparameters

The model consumes one irregular path per valid anchor row. Each path contains relative time plus meteorological and seasonal channels; the target total_ice_m is predicted, not fed back as an input feature.

Input Channels

1. relative_time
2. doy_sin
3. doy_cos
4. Ten_Meter_Elevation_Wind_Speed_meterPerSecond
5. Air_Temperature_celsius
6. Relative_Humidity_percent
7. Shortwave_Radiation_Downwelling_wattPerMeterSquared
8. Longwave_Radiation_Downwelling_wattPerMeterSquared
9. Sea_Level_Barometric_Pressure_pascal
10. Surface_Level_Barometric_Pressure_pascal
11. Precipitation_millimeterPerDay
12. Snowfall_millimeterPerDay
13. ice_prev_m
14. ice_prev_gap_days
15. ice_prev_available

Core Training Configuration

Field	Value
window.window_days	7
coeffs.interpolation	hermite
model.method	rk4
model.hidden_channels	48

Field	Value
model.hidden_hidden_channels	96
model.num_hidden_layers	2
train.batch_size	64
train.learning_rate	0.0002
train.weight_decay	0.0005
train.loss	huber
train.huber_delta	0.1
train.max_epochs	120
train.gradient_clip_norm	0.5
custom_split.val_fraction	0.1
custom_split.min_train_rows_per_lake	24
custom_split.min_val_rows_per_lake	8
custom_split.max_train_windows_per_lake	160
custom_split.max_val_windows_per_lake	20
custom_split.target_lake_test_start	2026-01-01

1. Source-Lake Selection, Windowing, And Balance Effects

Each source lake is first split by time into train/val. The target lake [Li] Lake Xiaoxingkai is split so that rows before the cutoff are further divided into train/val, while rows from the cutoff onward are reserved for test. A row only becomes a valid window if there are at least two observations inside the 7-day lookback. After that, train and val windows are capped per lake for balance.

Row Assignment Counts From The Source File

Lake	Rows In Source	Train Rows	Val Rows	Test Rows
[Li] Lake Xiaoxingkai	68	24	7	37
[Huo] Lake Hnagzhang	101	90	11	0
[Huo] Lake Ulansu	163	146	17	0
[IISD Experimental] Lake 239	160	144	16	0
[IISD Experimental] Lake 302	101	90	11	0
[North Temperate Lakes LTER] Big Muskellunge Lake	121	108	13	0
[North Temperate Lakes LTER] Bog 12-15 (Trout Bog)	124	111	13	0
[North Temperate Lakes LTER] Bog 27-2 (Crystal Bog)	121	108	13	0
[North Temperate Lakes LTER] Crystal Lake (NTL)	120	108	12	0

Window Eligibility And Balance Effects

Lake	Train Valid	Train Kept	Val Valid	Val Kept	Test Windows	Median Points
[Li] Lake Xiaoxingkai	23	23	7	7	36	8.0
[Huo] Lake Hnagzhang	89	89	11	11	0	8.0
[Huo] Lake Ulansu	144	144	17	17	0	8.0
[IISD Experimental] Lake 239	17	17	2	2	0	2.0
[IISD Experimental] Lake 302	5	5	3	3	0	2.0

Lake	Train Valid	Train Kept	Val Valid	Val Kept	Test Windows	Median Points
[North Temperate Lakes LTER] Big Muskellunge Lake	0	0	0	0	0	0.0
[North Temperate Lakes LTER] Bog 12-15 (Trout Bog)	0	0	0	0	0	0.0
[North Temperate Lakes LTER] Bog 27-2 (Crystal Bog)	0	0	0	0	0	0.0
[North Temperate Lakes LTER] Crystal Lake (NTL)	0	0	0	0	0	0.0

Total Counts

Total included source rows: 1079

Total train rows / val rows / test rows: 929 / 113 / 37

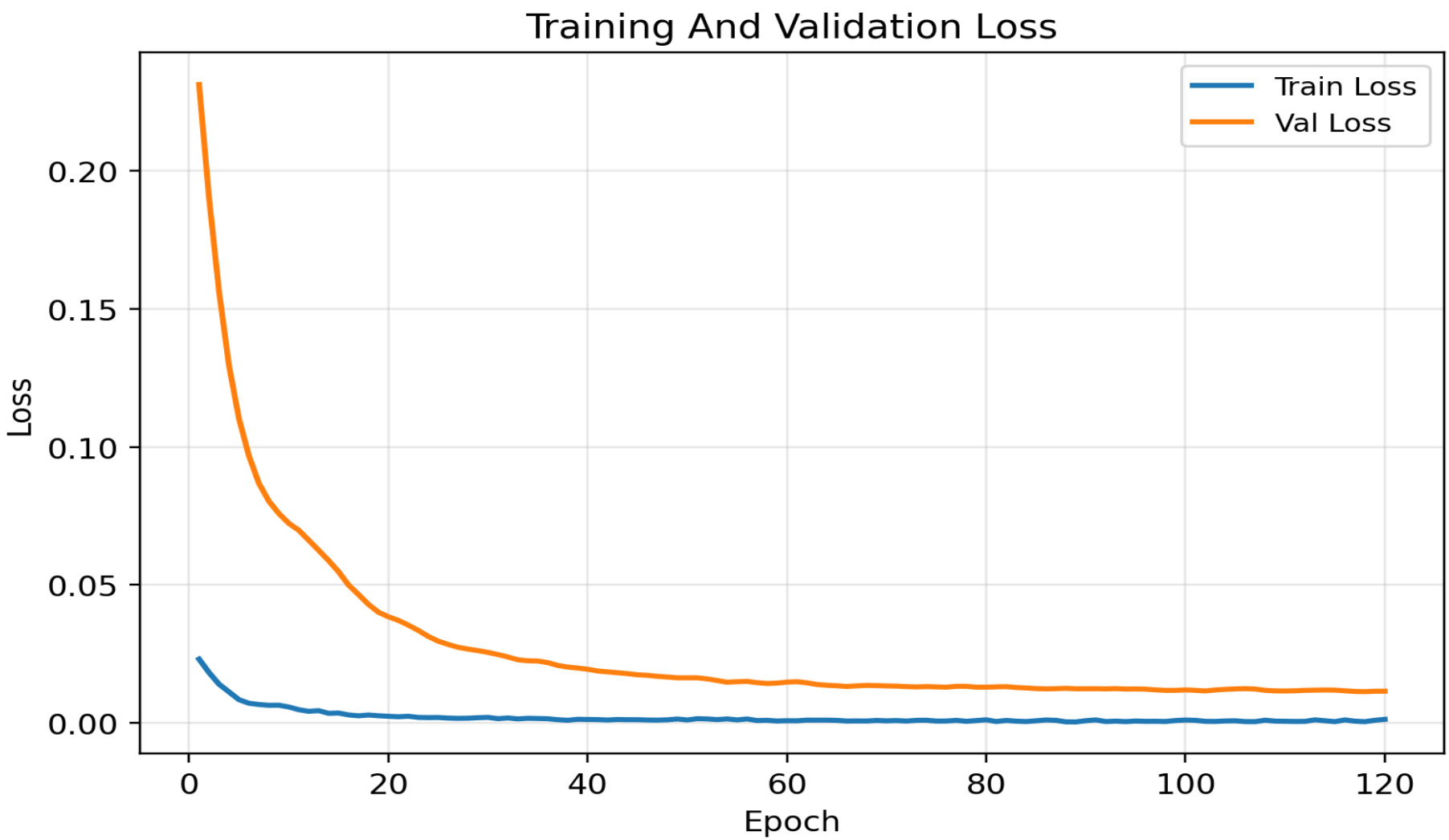
Total train windows before/after balance: 278 / 278

Total val windows before/after balance: 40 / 40

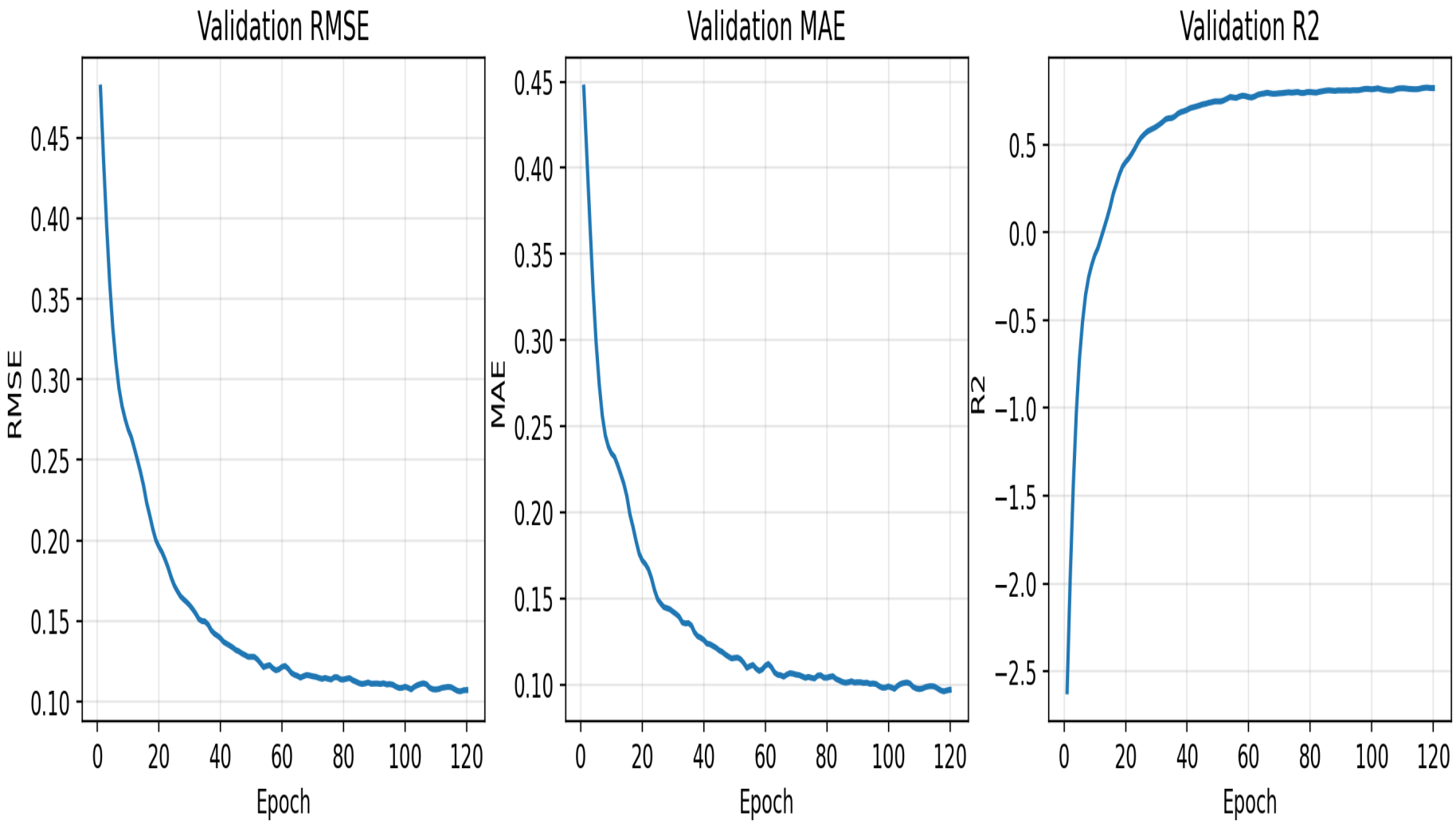
Total test windows: 36

2. Core Result Figures

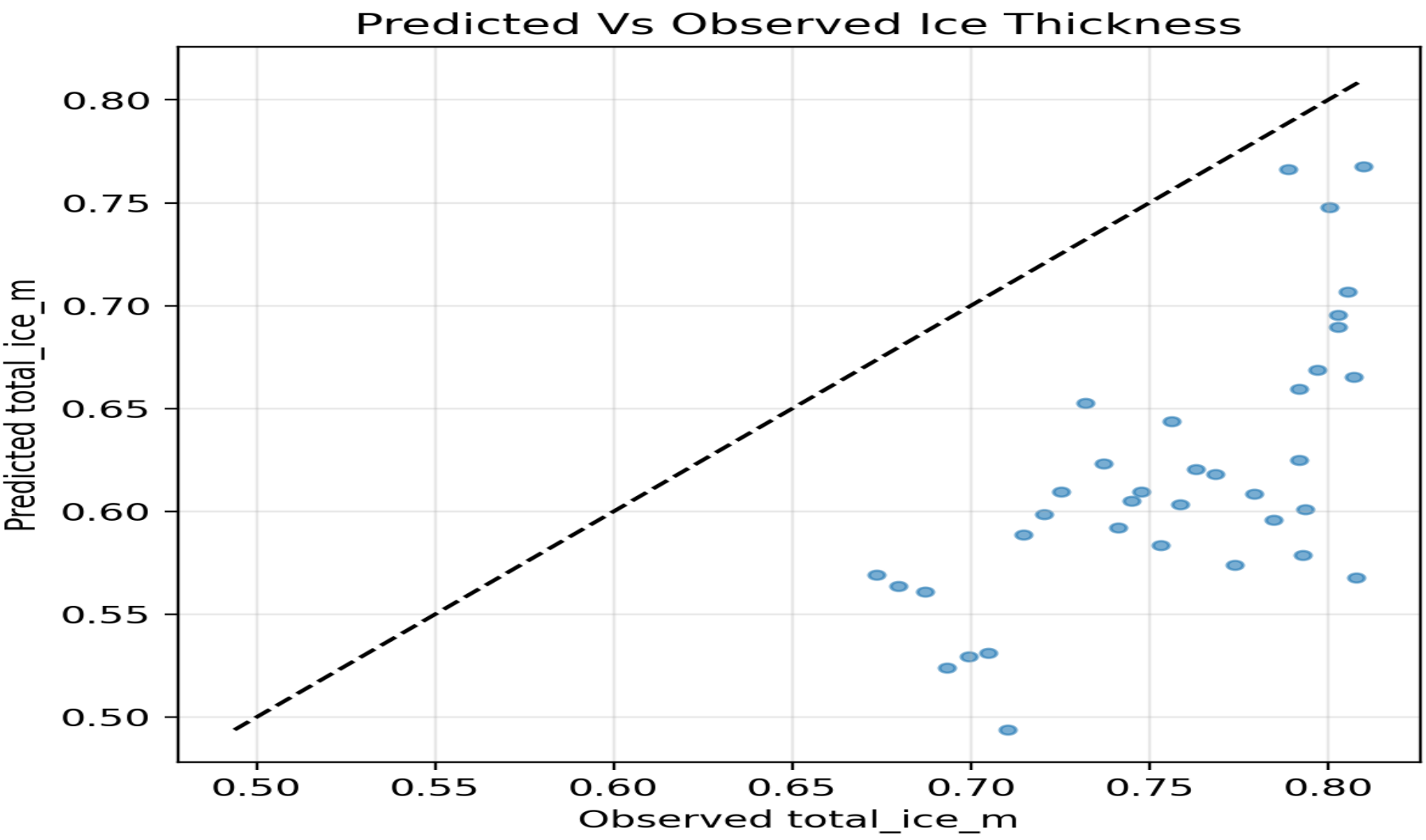
Training loss curve



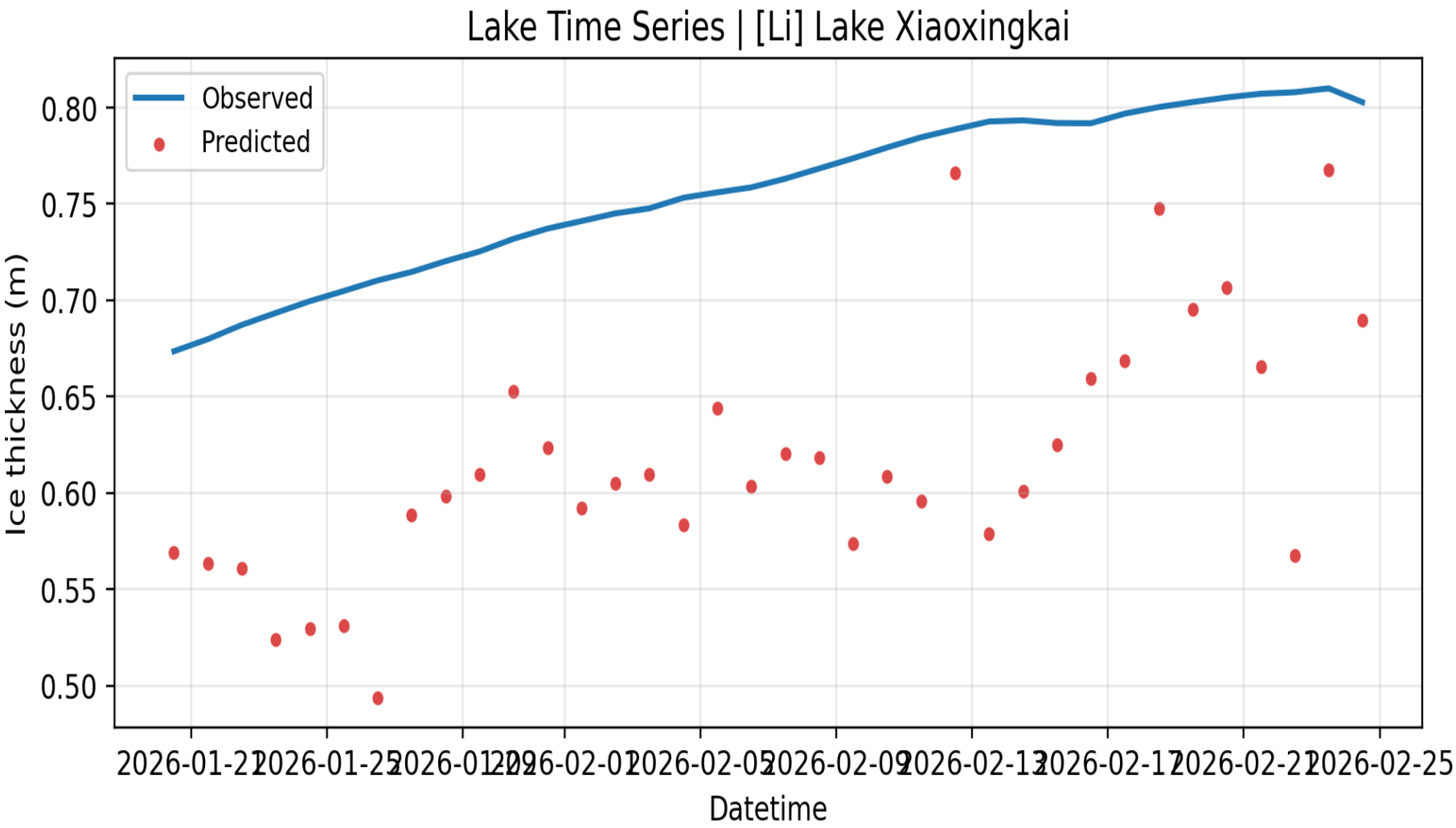
Validation metric curves



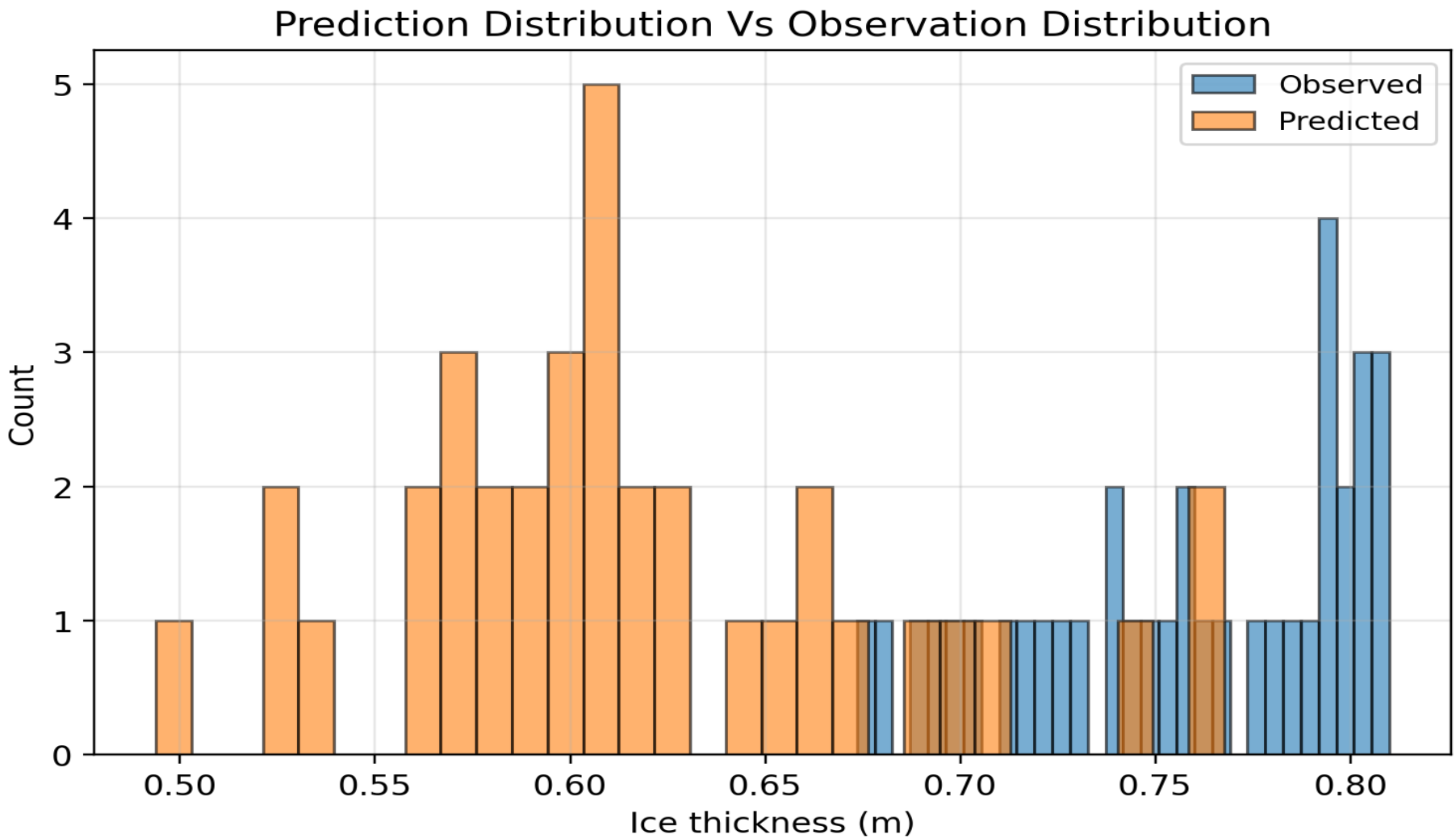
Evaluation predicted vs observed



Xiaoxingkai evaluation time series



Prediction vs observation distribution

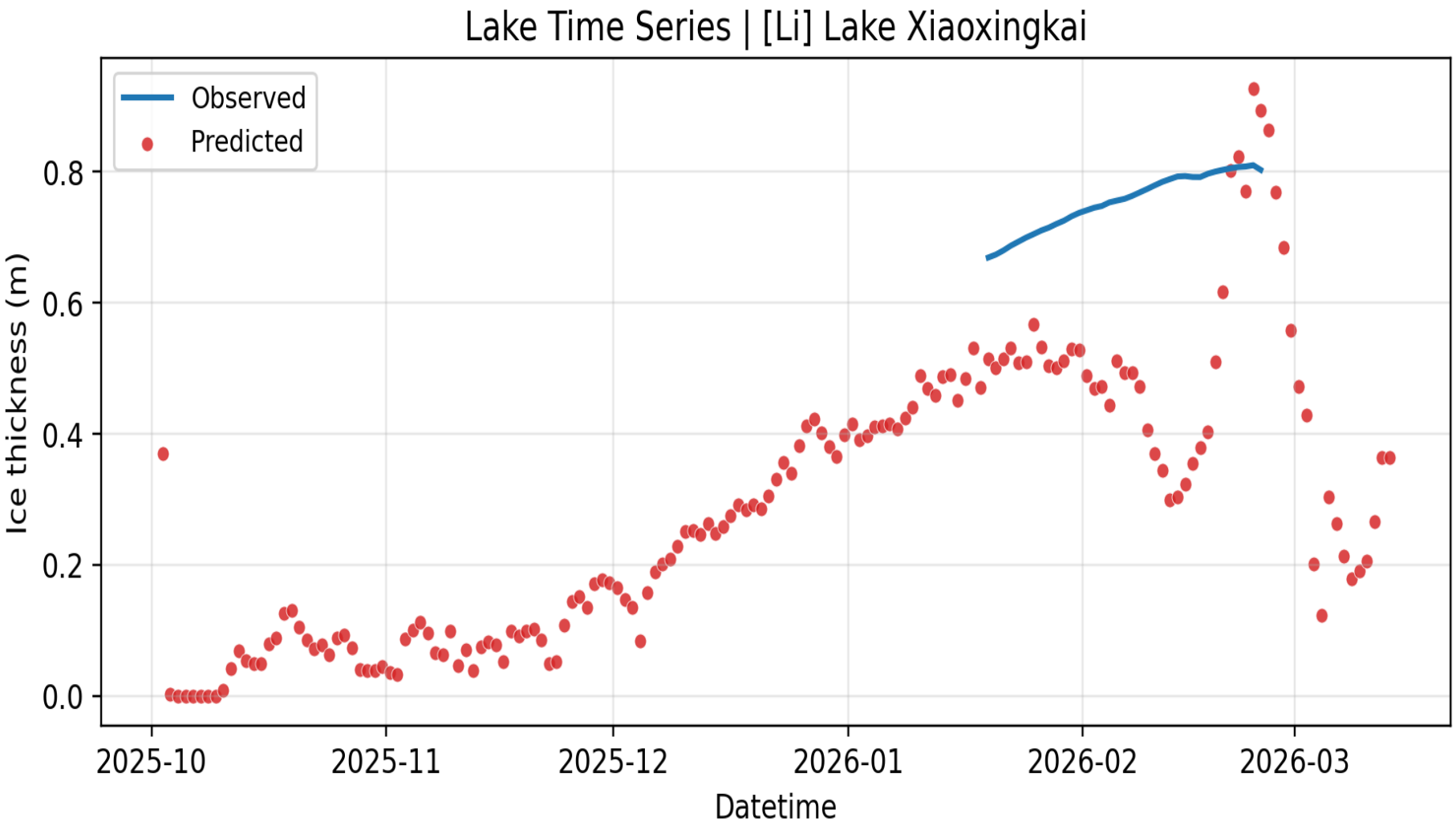

Prediction Distribution Vs Observation Distribution
Observed
Predicted
Count
0
1
2
3
4
5
0.50
0.55
0.60
0.65
0.70
0.75
0.80
Ice thickness (m)

3. Seasonal Rollout Test

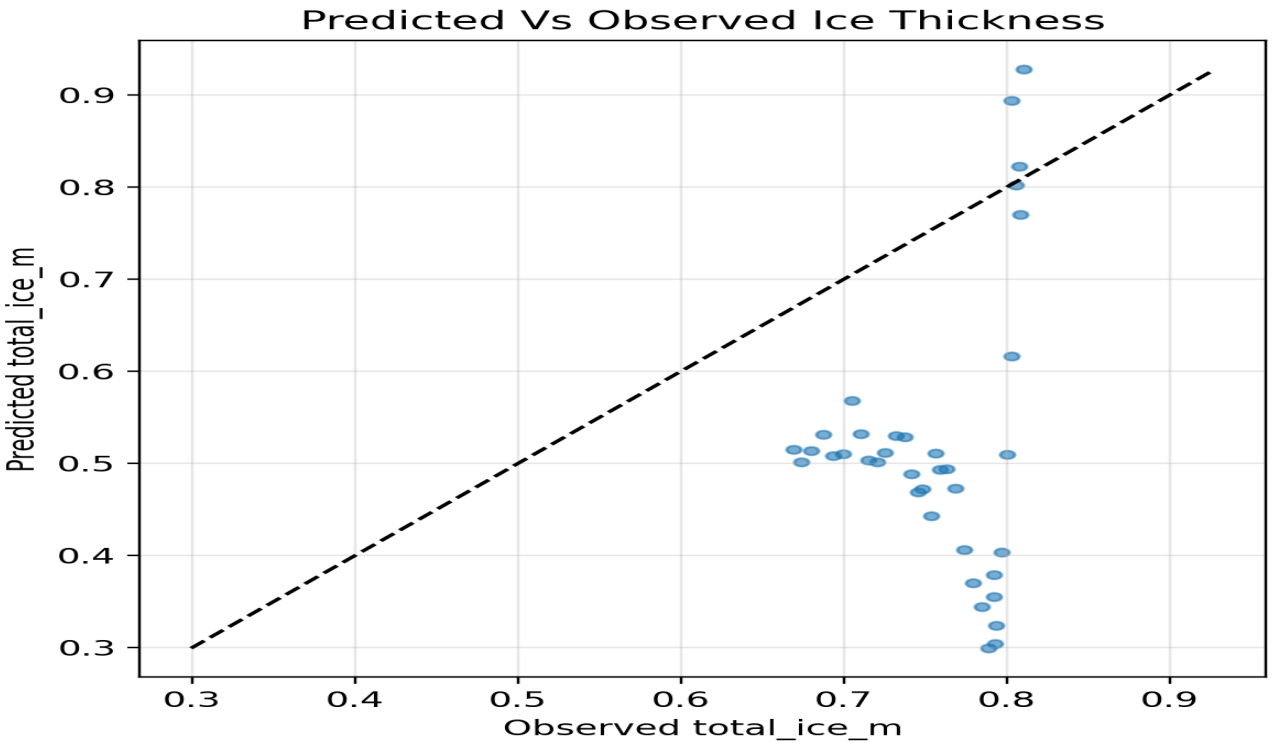
This second test reuses the same trained model, but drives it with Xiaoxingkai daily 12:00 ERA5 from the continuous hourly archive. It answers whether the model can follow the full ice-season evolution when given uninterrupted meteorological forcing rather than only the held-out observation rows.

Field	Value
count	37
rmse	0.2803
mae	0.2496
r2	-41.1819
bias	-0.2375
rollout_rows	163
overlap_rows	37
start_datetime	2025-10-02 12:00:00
end_datetime	2026-03-13 12:00:00

Seasonal rollout time series



Seasonal rollout overlap: predicted vs observed


Predicted Vs Observed Ice Thickness
Predicted total_ice_m
Observed total_ice_m
0.3
0.4
0.5
0.6
0.7
0.8
0.9